

Fig. 4: Design Calculators

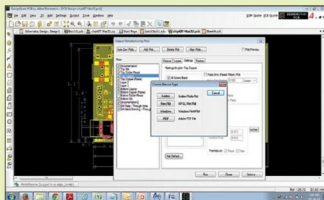


Fig. 5: Manufacturing plot



Fig. 6: 3D view

About DesignSpark PCB

Initial release: July 5, 2010

Stable release: Version 6.0 on January 21, 2014

Operating system: Windows, Linux

Type: Free EDA software

Licence: Proprietary software

Top competing tools

- > Altium Designer
- > Eagle
- > Fritzing
- > gEDA Project
- > KiCAD
- > Ni Circuit Design Suite
- > Proteus PCB Design
- > TINA

Forward Design Change and Back Annotations.

After translating schematic to PCB, if any changes are made in schematic the same can be transferred to PCB layout using Forward Design Changes option. It works to achieve netlist synchronisation between related schematic sheets and PCB designs. Design changes (addition/deletion) that are pushed forward include Net name, Connection/Net, Net class, Component, Component values, Component groups, Package, Pin changes.

Back Annotations let designers make changes in the PCB design and then apply those changes back to the master or original schematic design.

Manufacturing plot (Auto-generating plot). Auto-generating plot lets designers

automatically generate manufacturing plot. Auto-generation of plots for a design depends on the type of design and the type of output device selected. Designers may choose one of the following device types: Gerber, Pen Plot, Windows, PDF. Once the device type is chosen, designers may select what their set of plots should cover.

3D Viewer, DXF and IDF export.

DesignSpark PCB supports 3D view of the PCB, which lets designers actually analyse how their PCB would look

like physically. It uses a simple set of rules describing how each PCB symbol should be represented, together with various settings, like the thickness of the board, and uses the industry-standard OpenGL graphics toolkit to draw the 3D view.

In addition, designers can export their PCB in MCAD tool for designing product enclosure. DesignSpark PCB supports two MCAD export formats: DFX and IDF. DFX output generates an AutoCAD DXF format file containing the board outline shape and the requested drill holes. The IDF format is used as an alternative to DFX by some MCAD (mechanical CAD) vendors for importing into their own systems. IDF files are exported in IDF 3.0 format. The IDF format is used by DesignSpark Mechanical (free and professional 3D model design software sponsored by RS Components) for importing board and component data from PCB design.

Good tool with a good ecosystem

DesignSpark PCB has lots of technical resources available online to help designers easily learn and use the software. DesignSpark.com features video and text tutorials for designers to get started on DesignSpark PCB. There are also technical documentations on install and upgrade instructions, list of input/output file formats and lots of other information. A technical forum is also available on DesignSpark.com for specific technical clarifications and questions. Periodic releases of new versions of the software maximise designers' experience as new versions are released based on users' feedback and are available for free.

Whether you are a professional designer making money from your designs, an electronics education professional, a student or a hobbyist seeking an easy-to-use, professional but unrestricted schematic capture and PCB layout tool, DesignSpark PCB could be the right one for you. ●

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